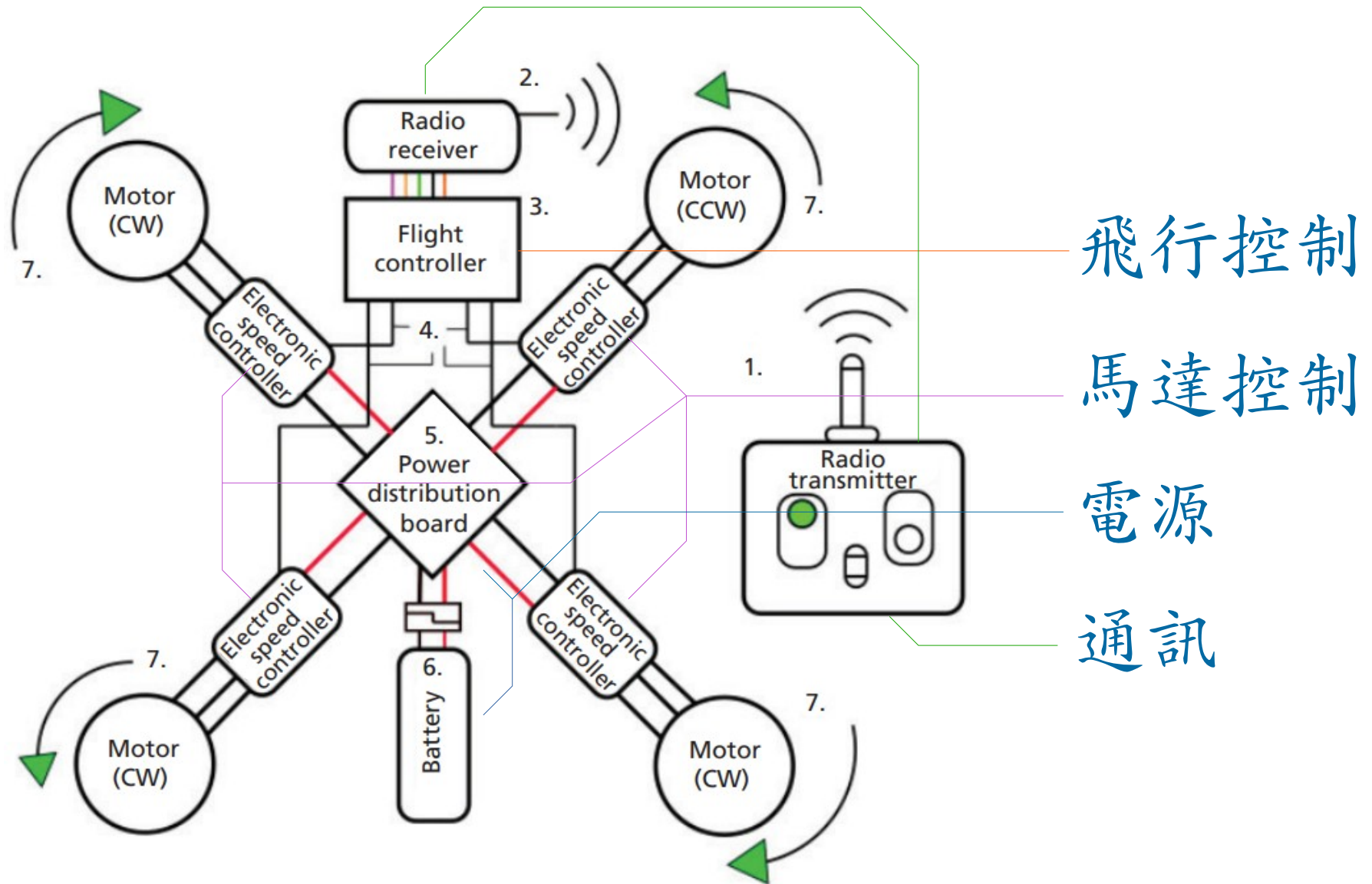


飛控板，電調板，電源架構 槳葉，電池

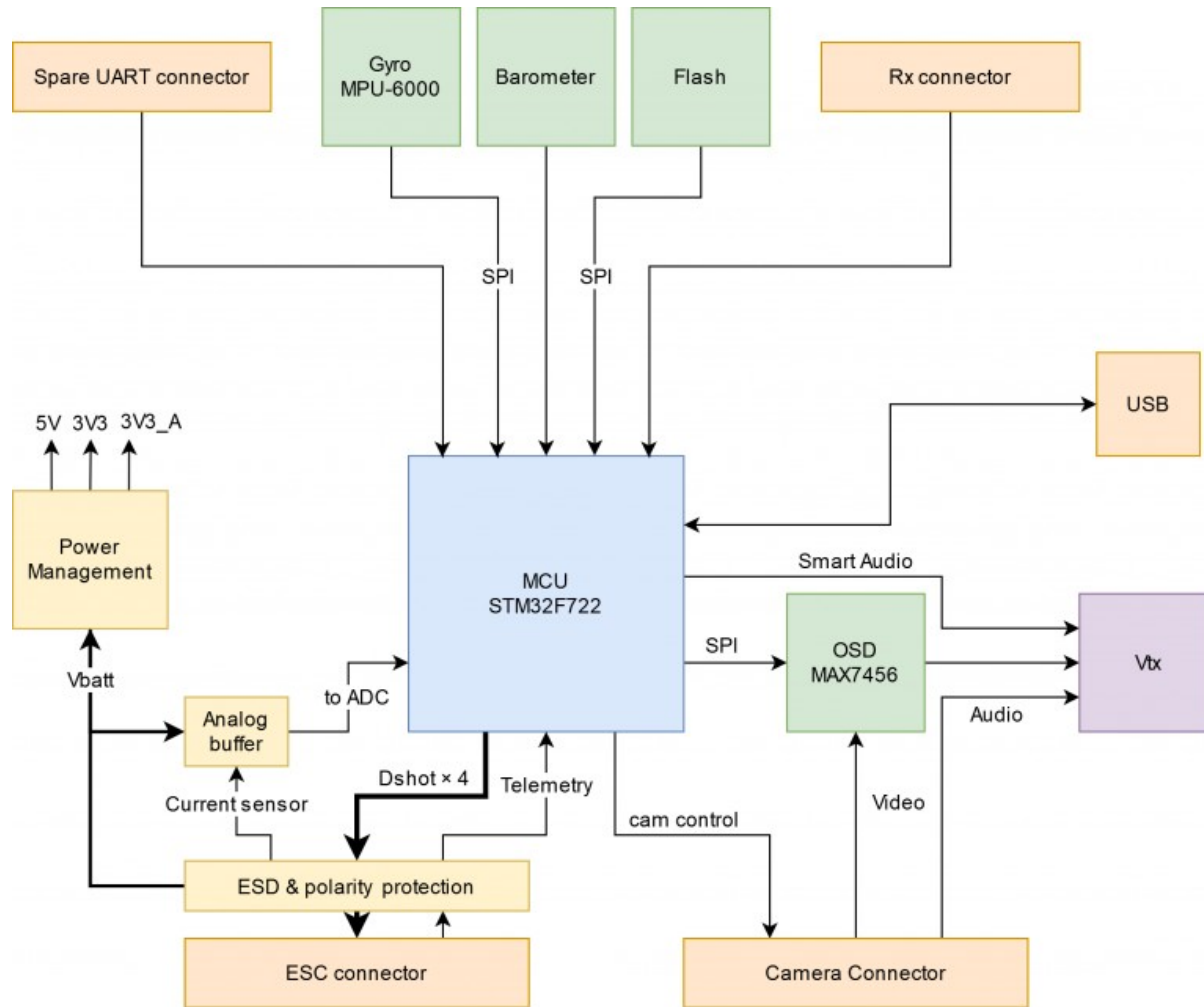
第六週
Nov/6/2026

中原大學電機系

四軸無人機電子系統



飛行控制電路板架構



MCU 處理器

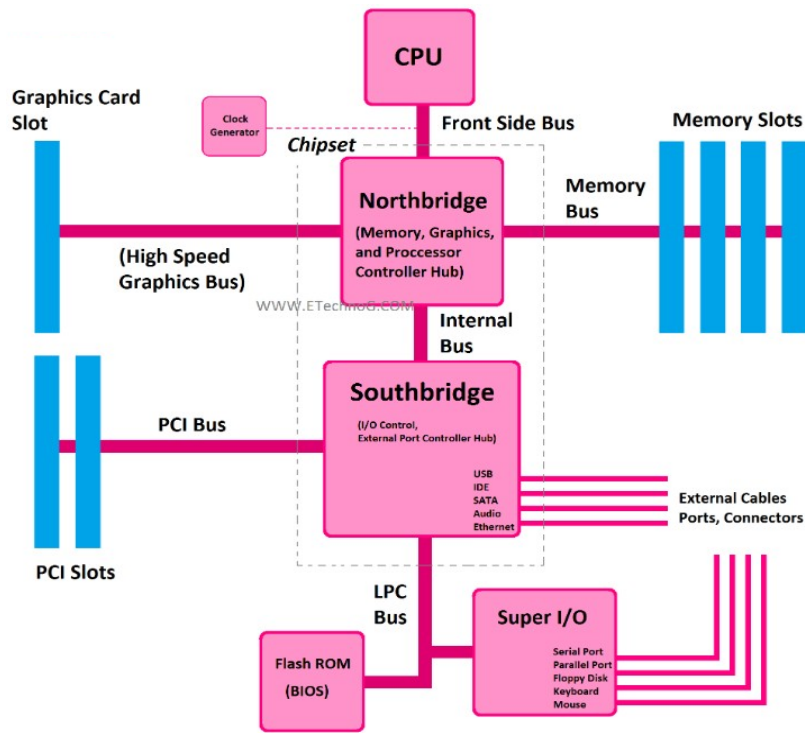
感測器

電源管理

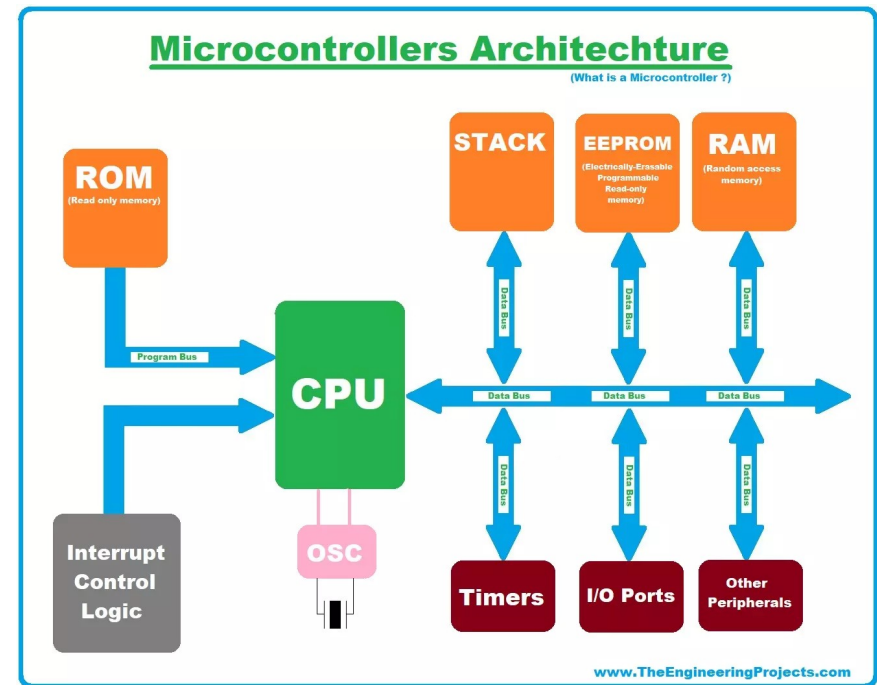
影資整合

界面聯接器

飛行控制電路板 - MCU

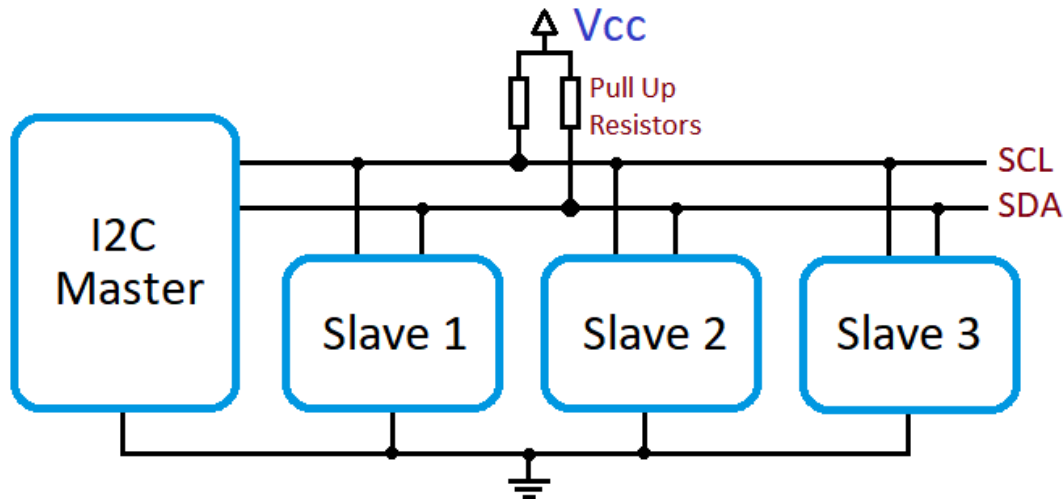


電腦主機板架構，中央處理器為獨立單一晶片。需要作業系統來同時處理不同的工作。



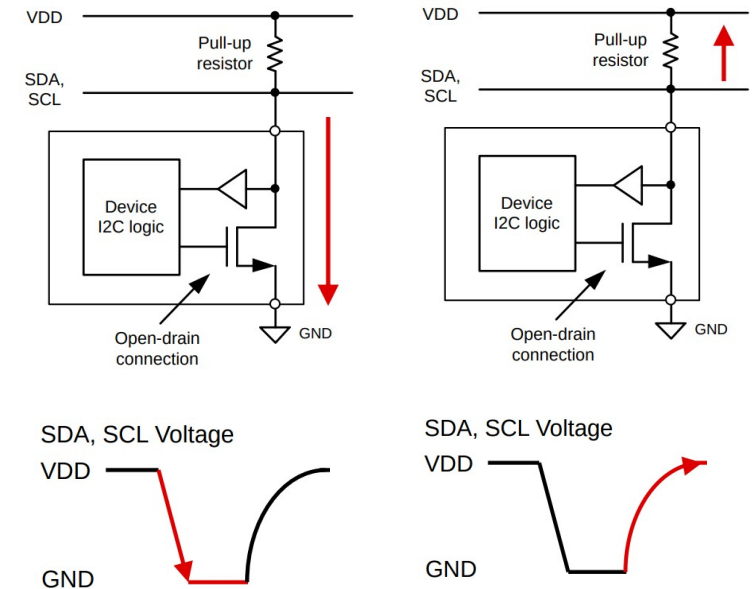
微控制器整合中央處理器和其他週邊於單一晶片，不需要作業系統處理特定工作。

MCU 與週邊通訊界面 - I2C

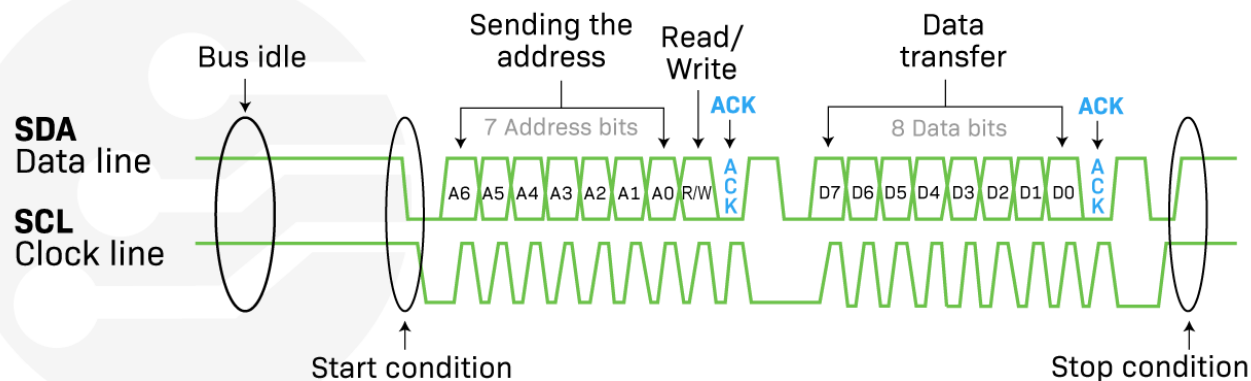


<https://justdoelectronics.com/understand-i2c-communication-protocol/>

Open Drain



I²C Data Communication Process

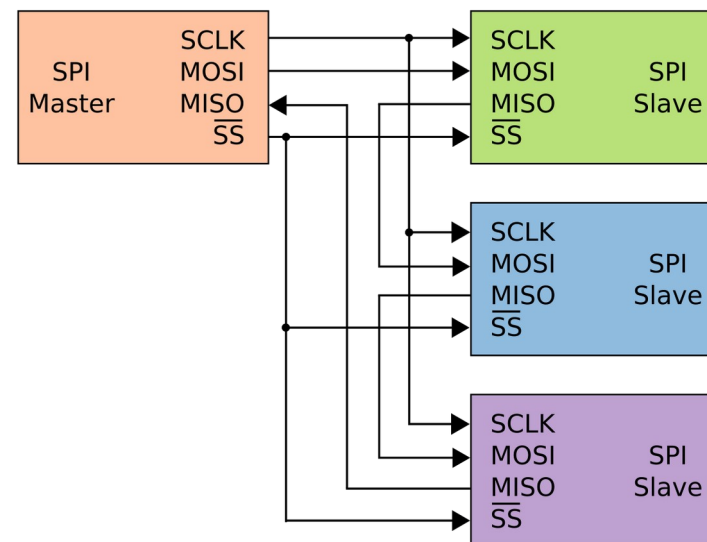
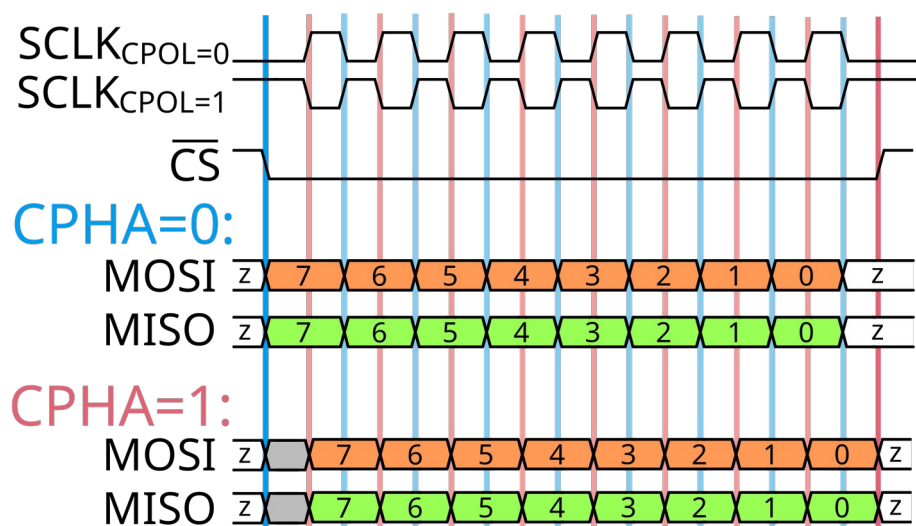


<https://newhavendisplay.com/blog/i2c-communication-interface/>

I2C 速度中等
但資訊傳輸有保障
利用定址指定通訊對象

MCU 與週邊通訊界面 - SPI

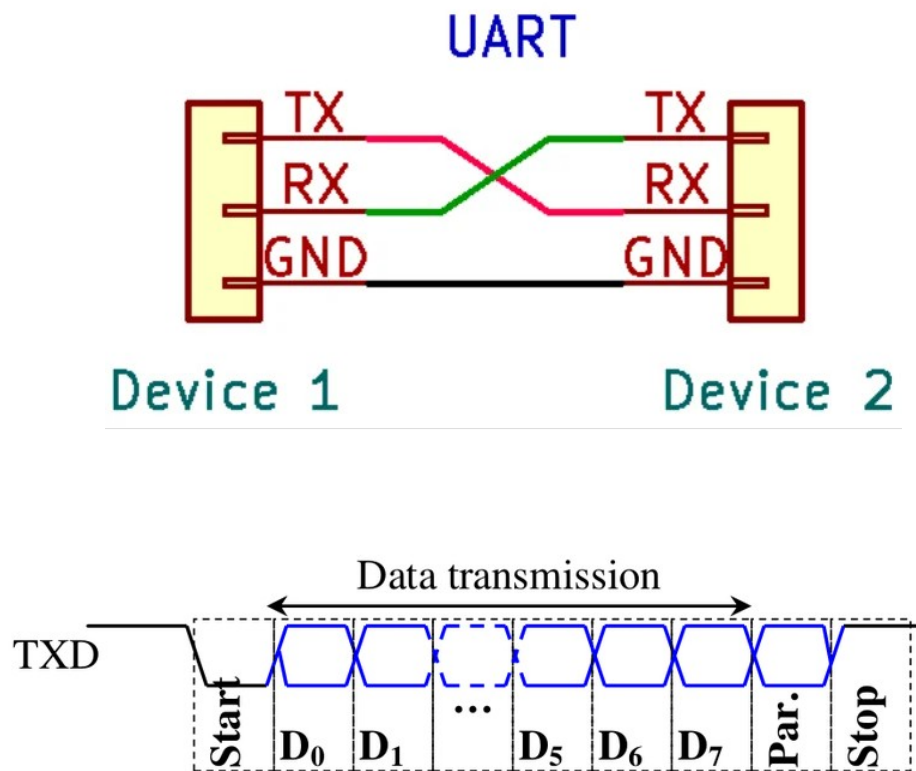
SPI 四種設定模式



SPI Mode	Clock Polarity	Clock Phase	Data is Latched at	Data is shifted at
0	0	0	Rising Edge	Falling Edge
1	0	1	Falling Edge	Rising Edge
2	1	0	Rising Edge	Falling Edge
3	1	1	Falling Edge	Rising Edge

SPI 速度快，沒有 ACK 確保，雙向同時傳輸資訊，訊號線較多，利用 CS 訊號指定通訊對象。

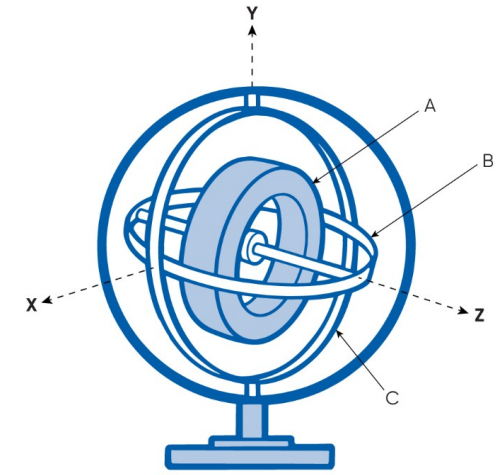
MCU 與週邊通訊界面 - UART



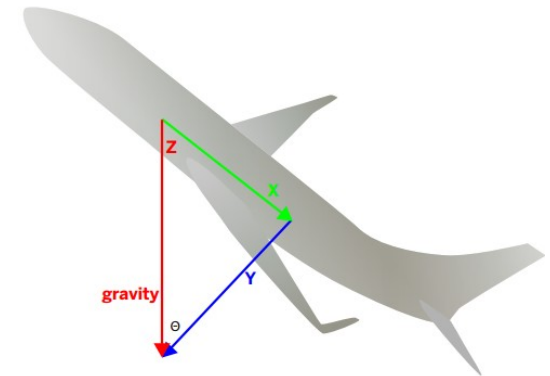
UART 是點對點通訊，可以設定傳輸速度，不需要時鐘訊號同步但兩個通訊元件傳輸速度要一致。使用 parity bit 來確保資訊完整。常用在 MCU 和 PC 間通訊，也會用在 MCU 和 GPS，藍芽等 MCU/週邊之間的通訊。

飛控板姿態感測器 - IMU

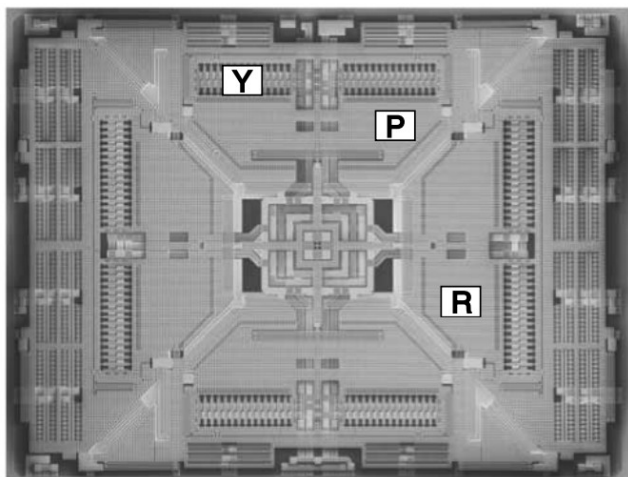
陀螺儀 偵測轉子方向的變化來測量角速度。當陀螺儀旋轉時，轉子具有角動量。如果施加外力試圖改變陀螺儀的方向，陀螺儀將因角動量守恆而抵抗這種變化。在 MEMS 中，此阻力產生扭矩，可以測量該扭矩以確定陀螺儀的角速度。



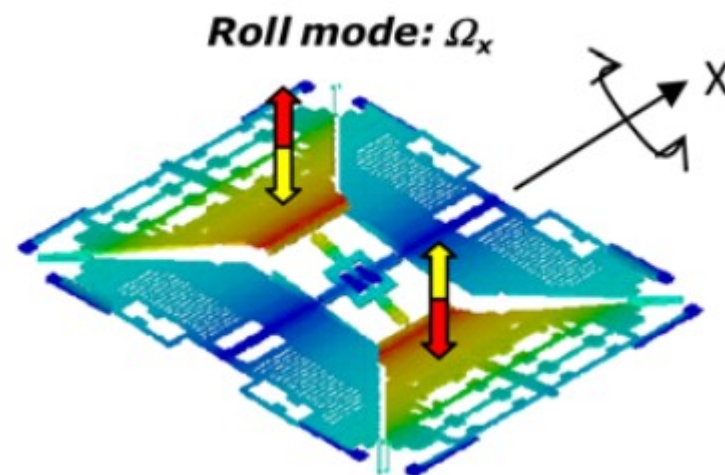
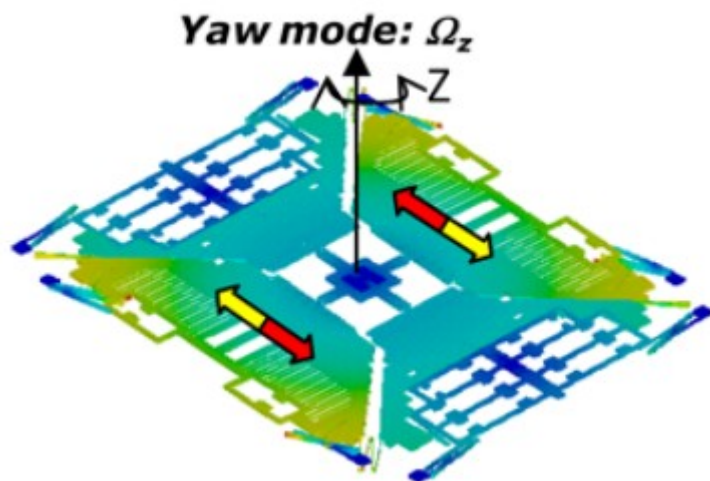
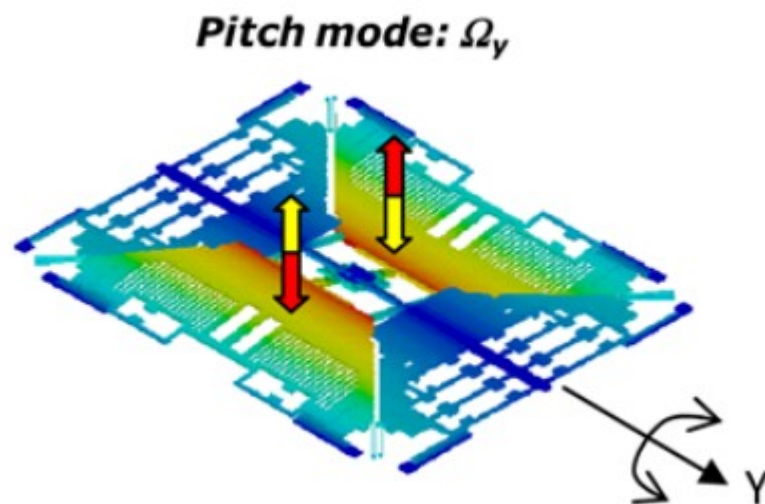
加速度計 可以透過測量地球重力加速度來偵測裝置的方向（傾斜），地球重力是作用在所有物體上的恆定向下力。



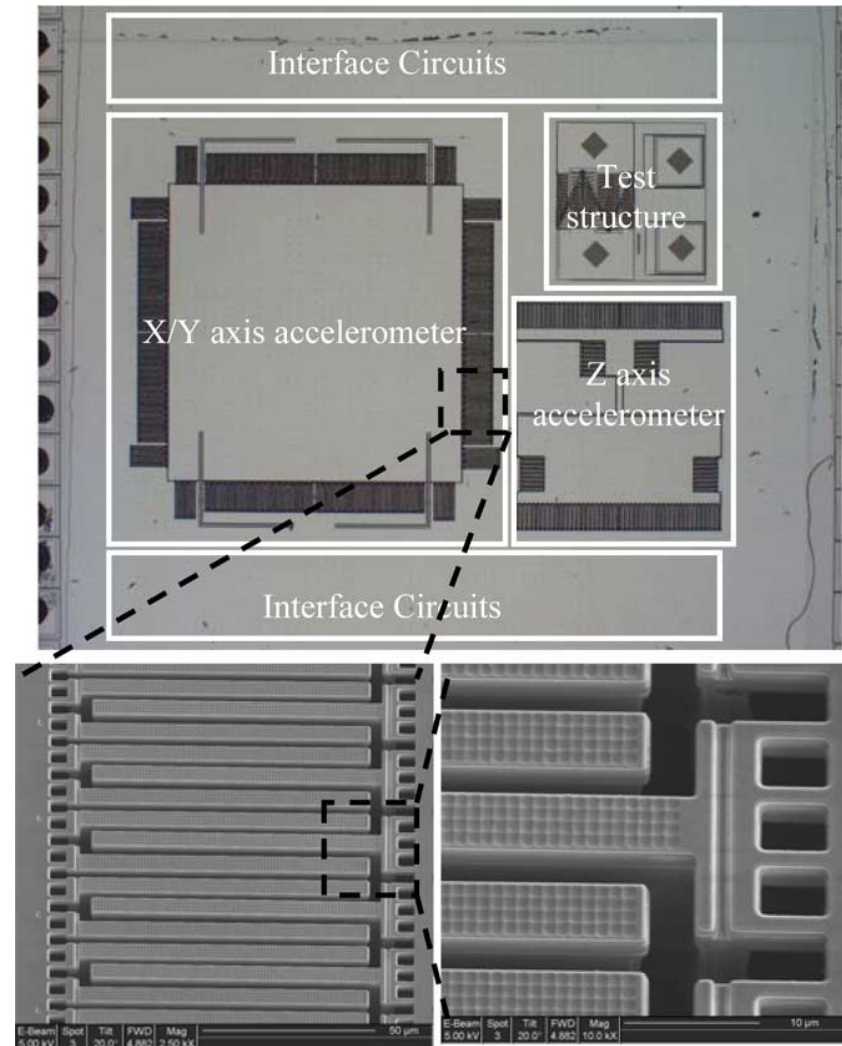
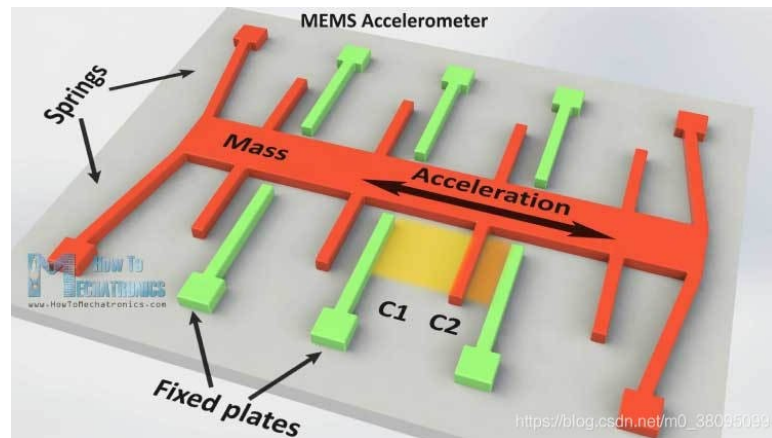
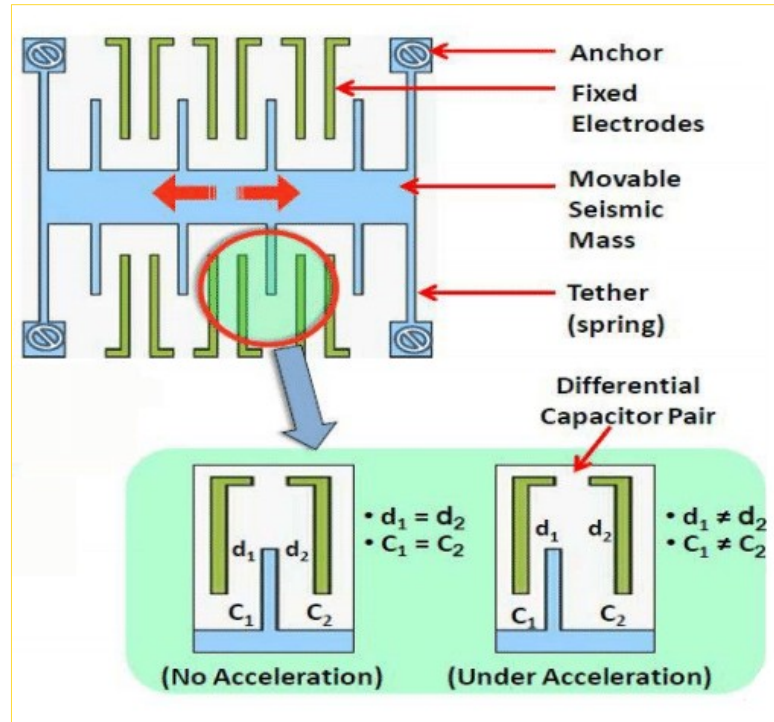
陀螺儀 - MEMS



MEMS structure die of 3-axis digital gyroscopes



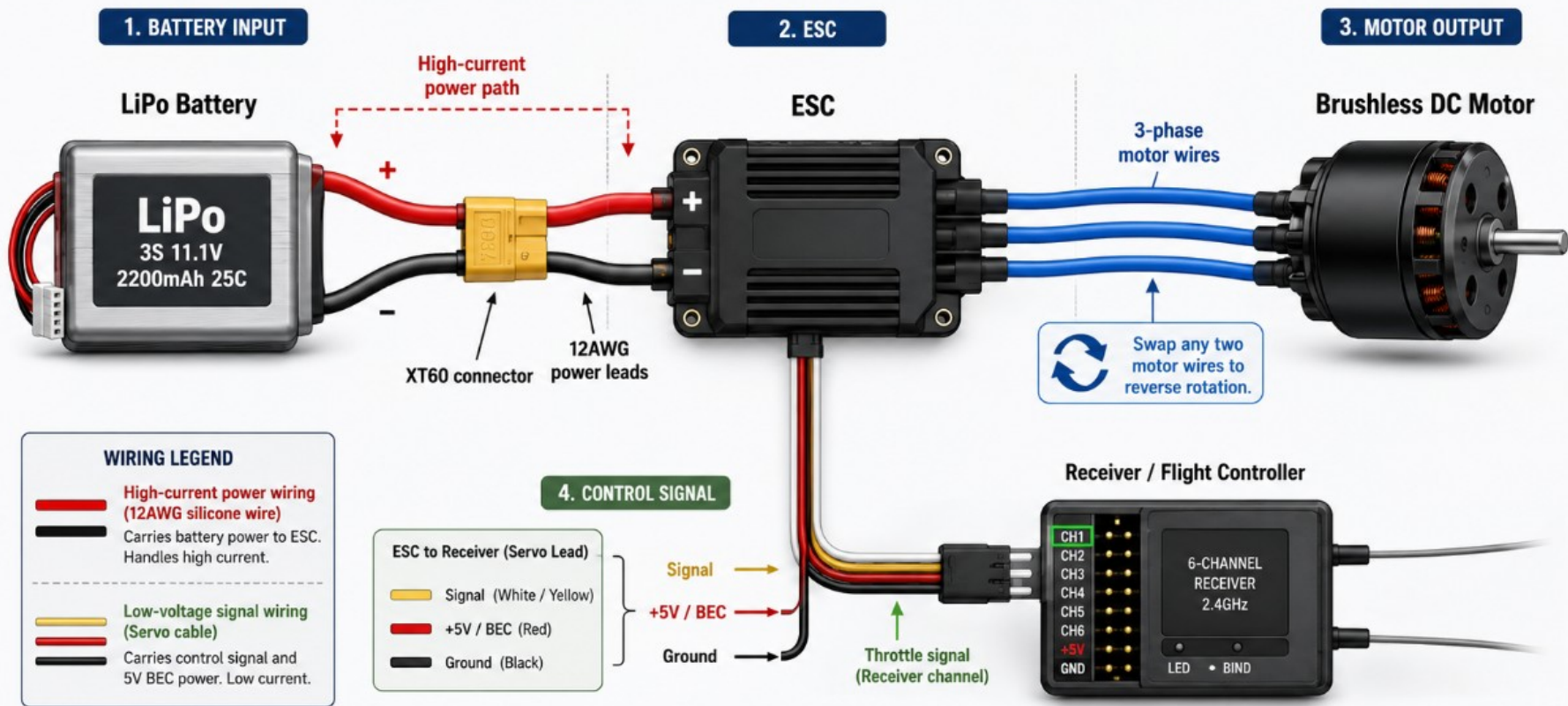
加速計 - MEMS



馬達控制 (Electronic Speed Controller)

ESC WIRING GUIDE – LIPO BATTERY TO MOTOR & RECEIVER

Basic wiring for a brushless ESC with LiPo battery, brushless motor and receiver / flight controller.

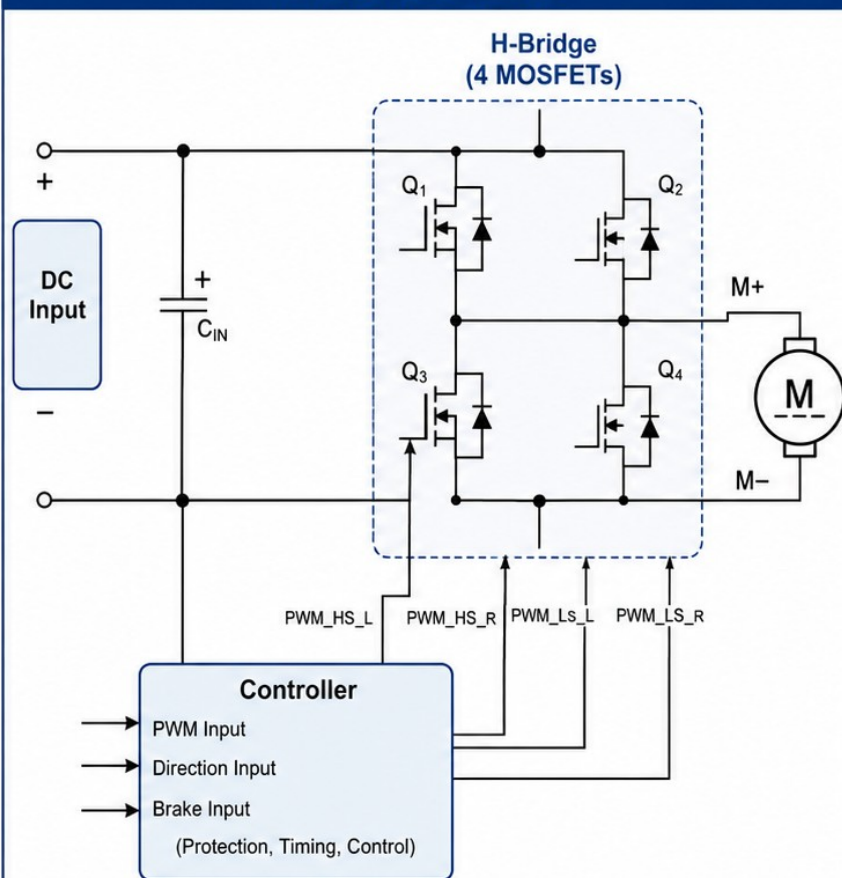


SAFETY NOTES: Always ensure correct polarity on power connections. Reversing + and - can damage the ESC.

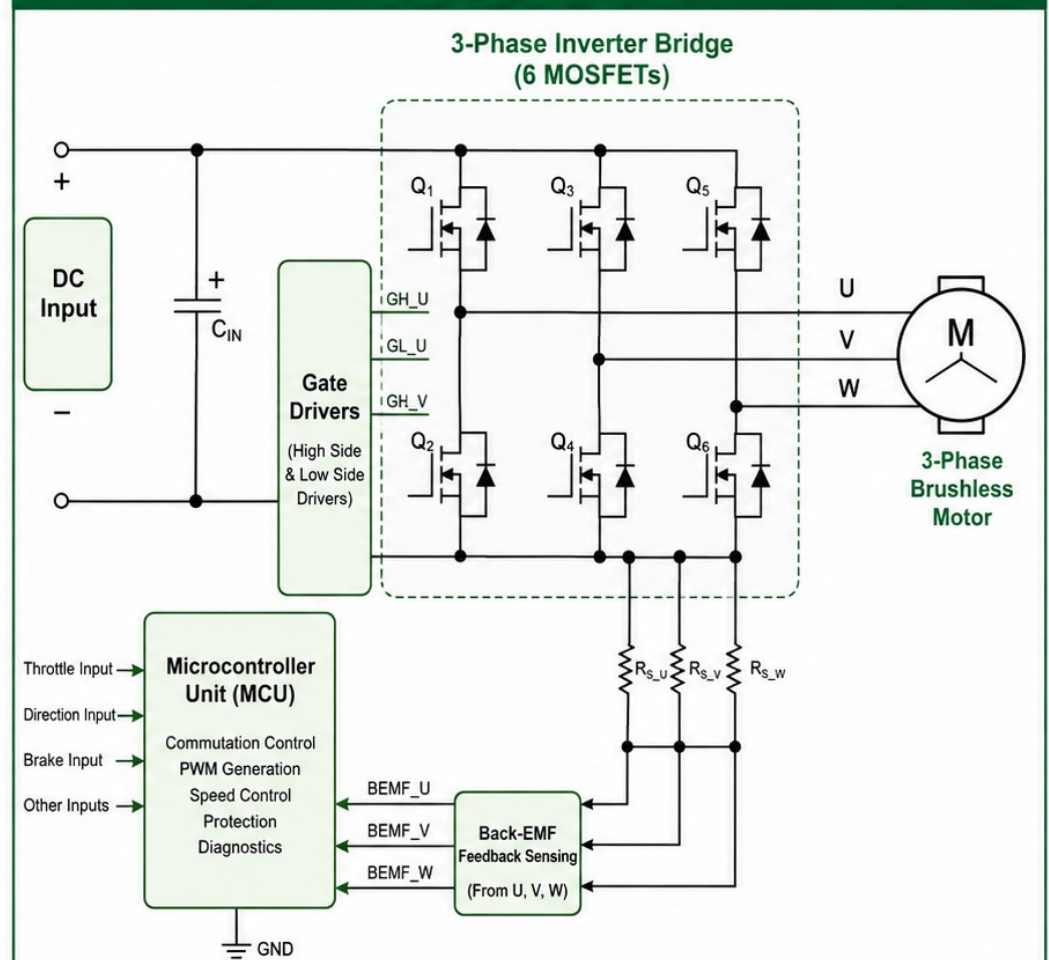
Ensure all connections are secure and well insulated. Do not short any wires.

馬達控制電路板架構

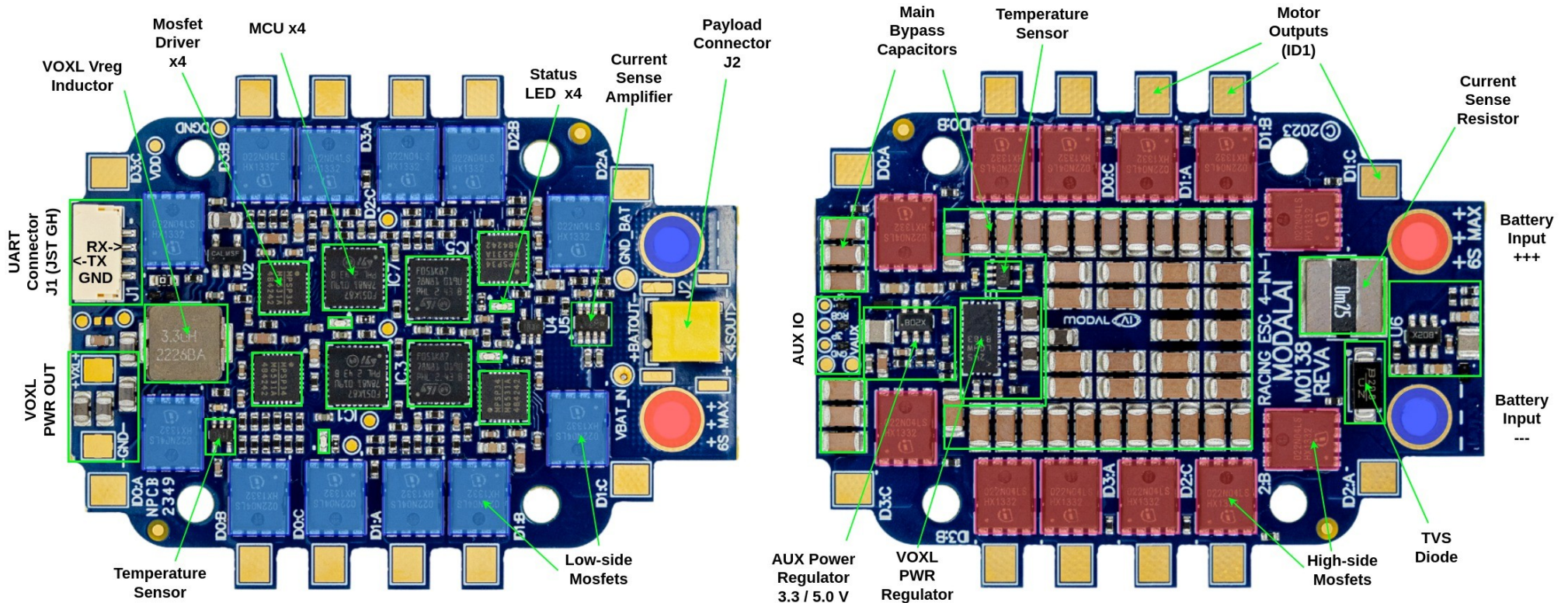
Brushed ESC



Brushless ESC



四合一馬達控制電路板

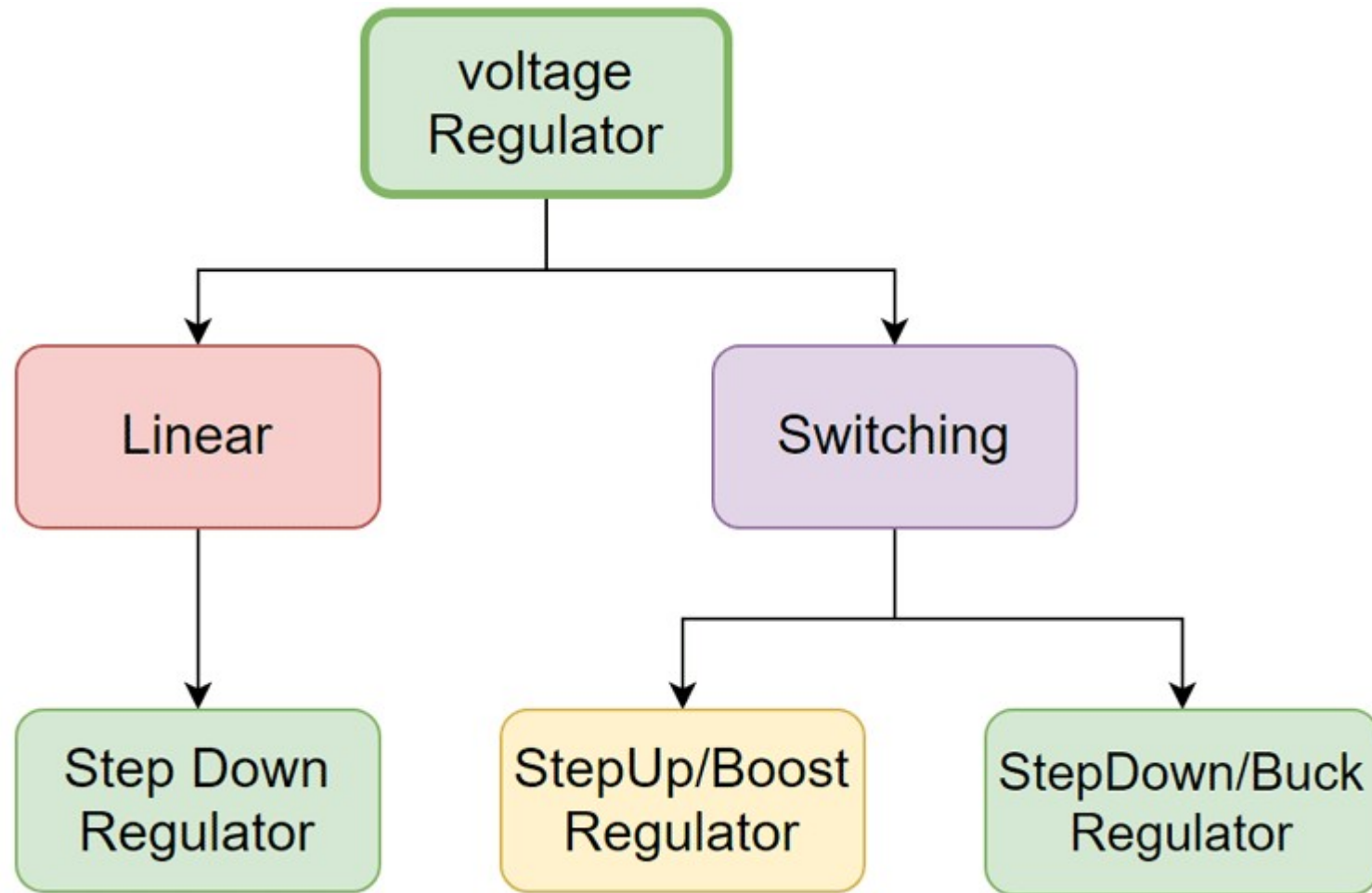


MCU x 4

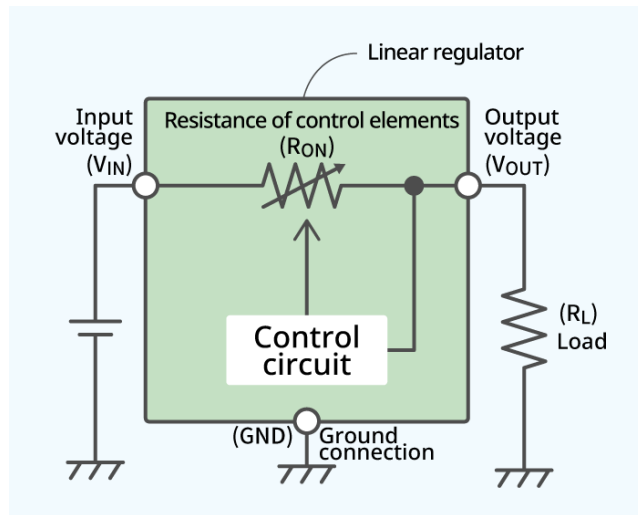
Gate Driver x 4

MOSFET x 24

電壓調節元件



電壓調節元件 - Low DropOut (LDO)



● Operation of error amplifier

The amplifier compares the reference voltage (V_{REF}) to the feedback voltage (V_{FB}) to control the on-resistance of the output driver

If V_{OUT} = the required value,

$$V_{REF} = V_{FB}$$

Ideal state

If V_{OUT} > the required value,

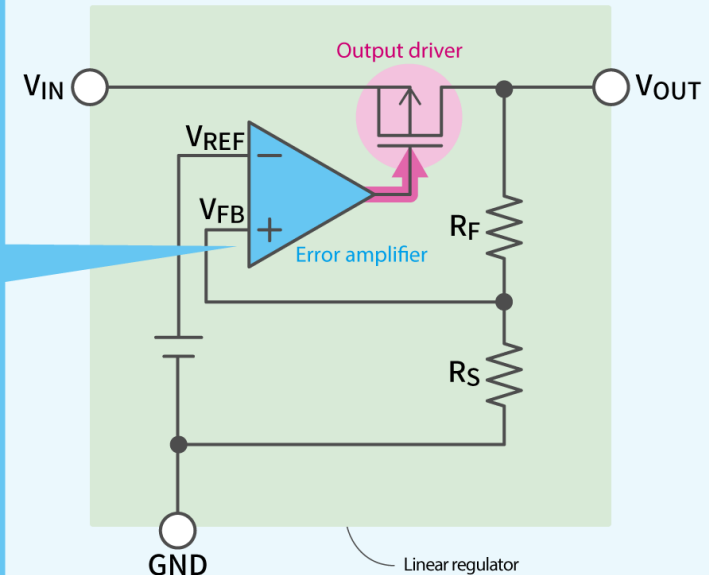
$$V_{REF} < V_{FB}$$

raises output driver on-resistance

if V_{OUT} < the required value,

$$V_{REF} > V_{FB}$$

lowers output driver on-resistance



<https://www.ablic.com/en/semicon/products/power-management-ic/voltage-regulator-ldo/intro-2/>

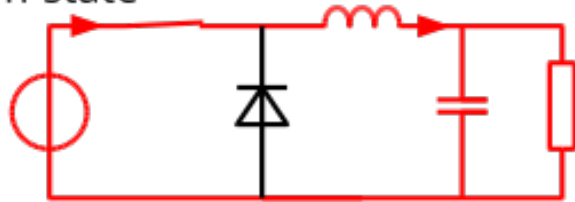
<https://www.ti.com/download/trng/docs/seminar/Topic%209%20-%20Understanding%20LDO%20dropout.pdf>

參考電壓如何產生：<https://hackmd.io/@chendylee/HyW-27Bc3>

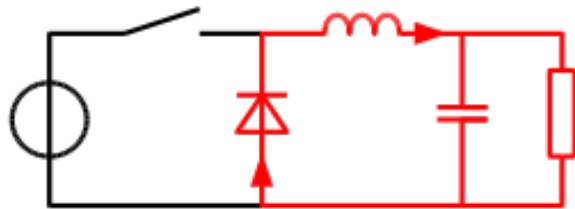
OpAMP 如何運作：<https://www.ti.com/lit/an/sboa092b/sboa092b.pdf>

電壓調節元件 - Buck Converter

On-state



Off-state



$$V_L = L \frac{dI_L}{dt}$$

$$\Delta I_{L_{on}} = \int_0^{t_{on}} \frac{V_L}{L} dt = \frac{V_{in} - V_{out}}{L} t_{on}$$

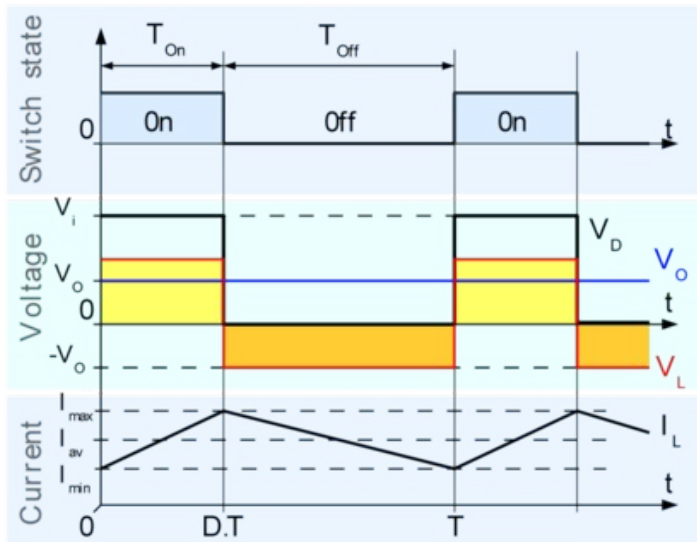
$$\Delta I_{L_{off}} = \int_{t_{on}}^T \frac{V_L}{L} dt = -\frac{V_{out}}{L} t_{off}$$

$$t_{on} = DT$$

$$t_{off} = (1 - D)T$$

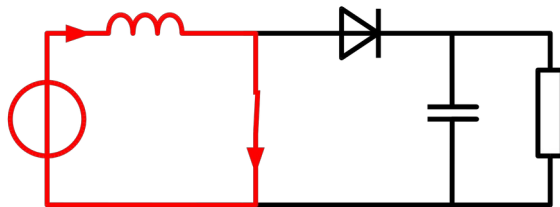
$$\Delta I_{L_{on}} + \Delta I_{L_{off}} = 0$$

$$\frac{V_{out}}{V_{in}} = D$$

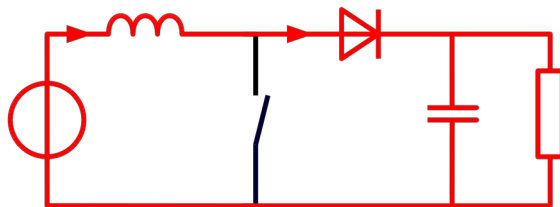


電壓調節元件 - Boost Converter

On-state

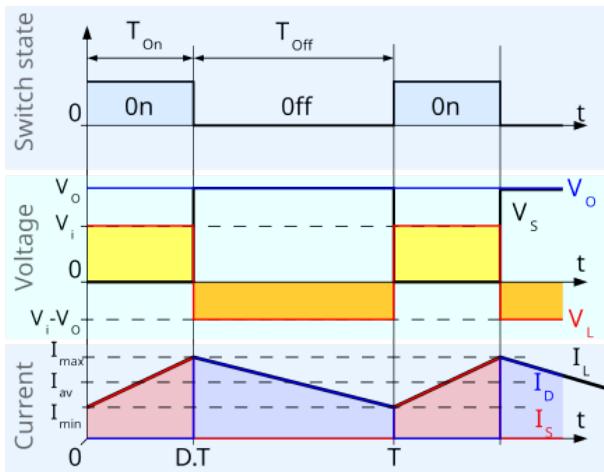


Off-state



$$\Delta I_{L_{on}} = \frac{1}{L} \int_0^{DT} V_{in} dt = \frac{DT}{L} V_{in}$$

$$\begin{aligned} \Delta I_{L_{off}} &= \int_{DT}^T \frac{V_{in} - V_{out}}{L} dt \\ &= \frac{(V_{in} - V_{out})(1 - D)T}{L} \end{aligned}$$

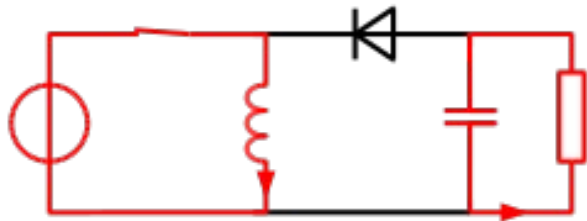


$$\Delta I_{L_{on}} + \Delta I_{L_{off}} = 0$$

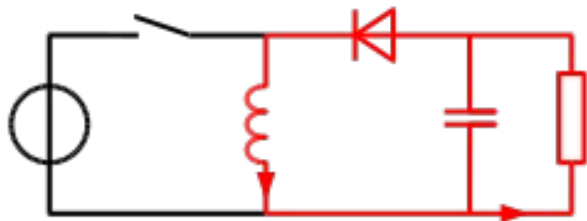
$$\frac{V_{out}}{V_{in}} = \frac{1}{1 - D}$$

電壓調節元件 - Buck-Boost Converter

On-state



Off-state



$$\Delta I_{L_{on}} = \int_0^{DT} dI_L$$

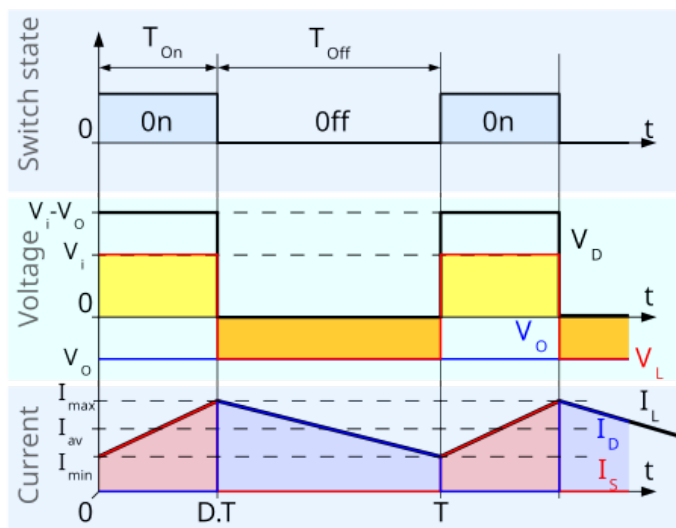
$$= \int_0^{DT} \frac{V_{in}}{L} dt = \frac{V_{in}DT}{L}$$

$$\Delta I_{L_{off}} = \int_0^{(1-D)T} dI_L$$

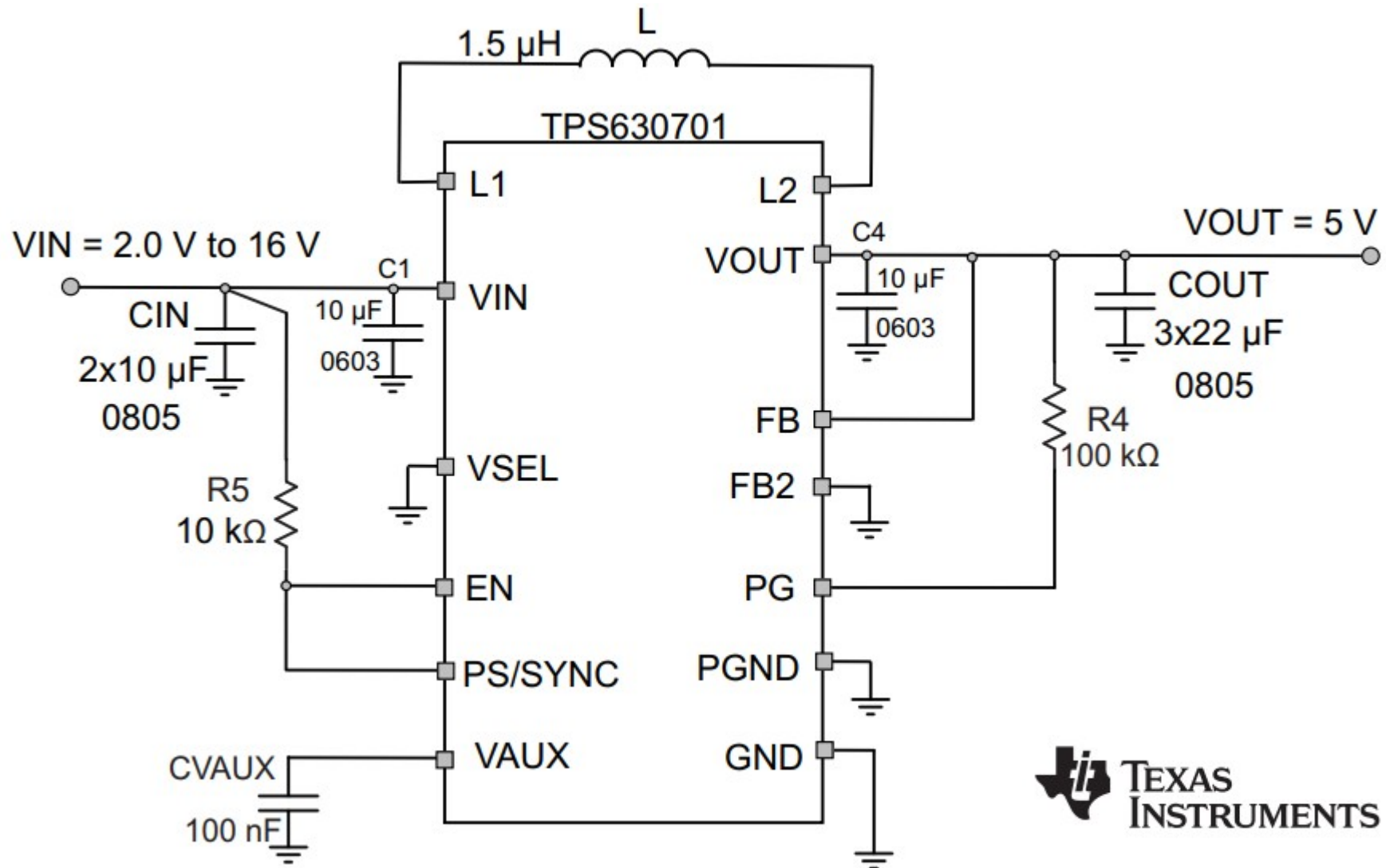
$$= \int_0^{(1-D)T} \frac{V_{out}}{L} dt = \frac{V_{out}(1-D)T}{L}$$

$$\Delta I_{L_{on}} + \Delta I_{L_{off}} = 0$$

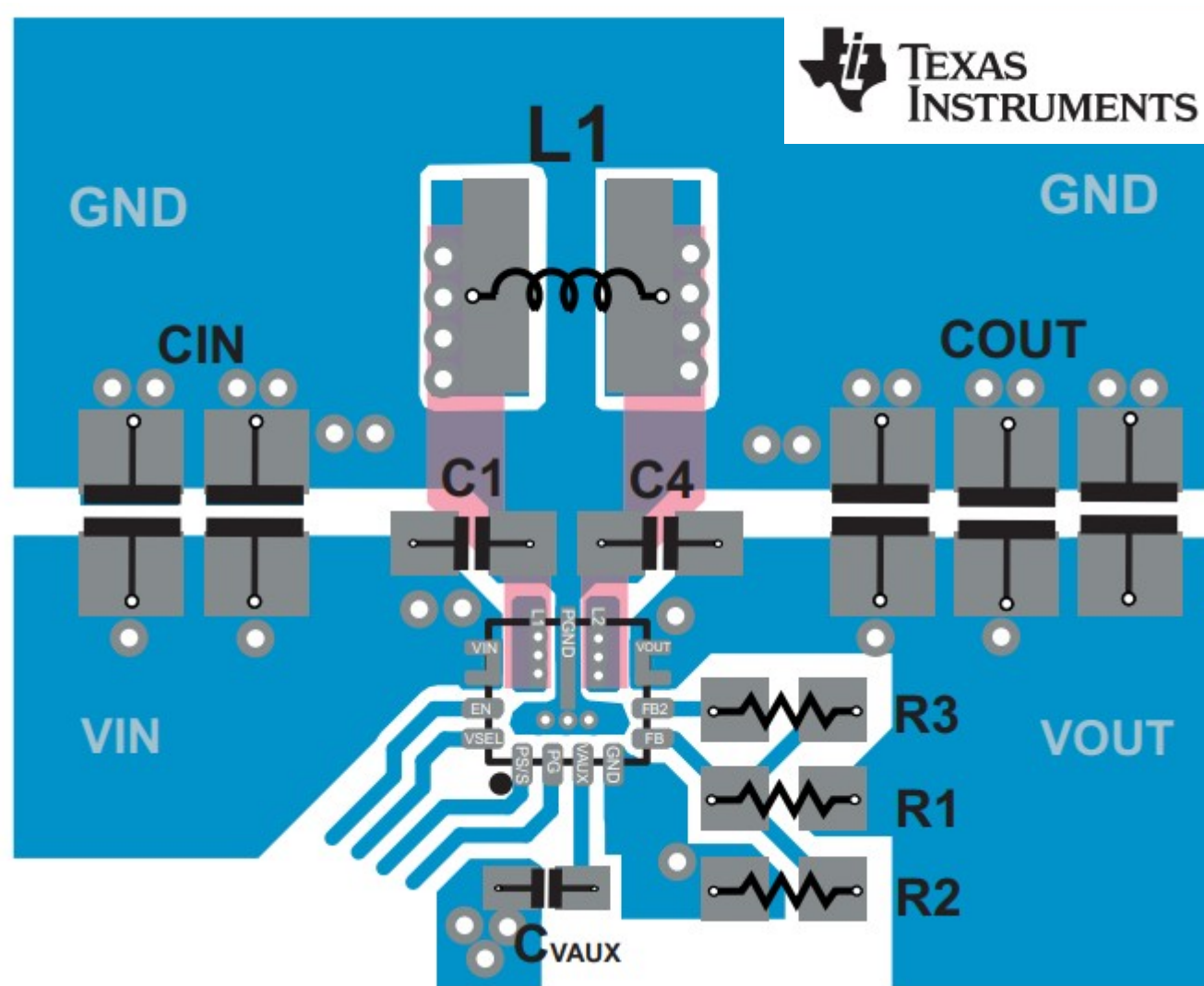
$$D = \frac{V_{out}}{V_{out} - V_{in}}$$



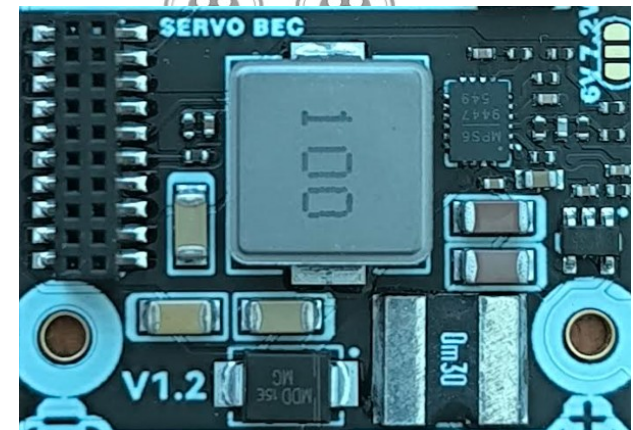
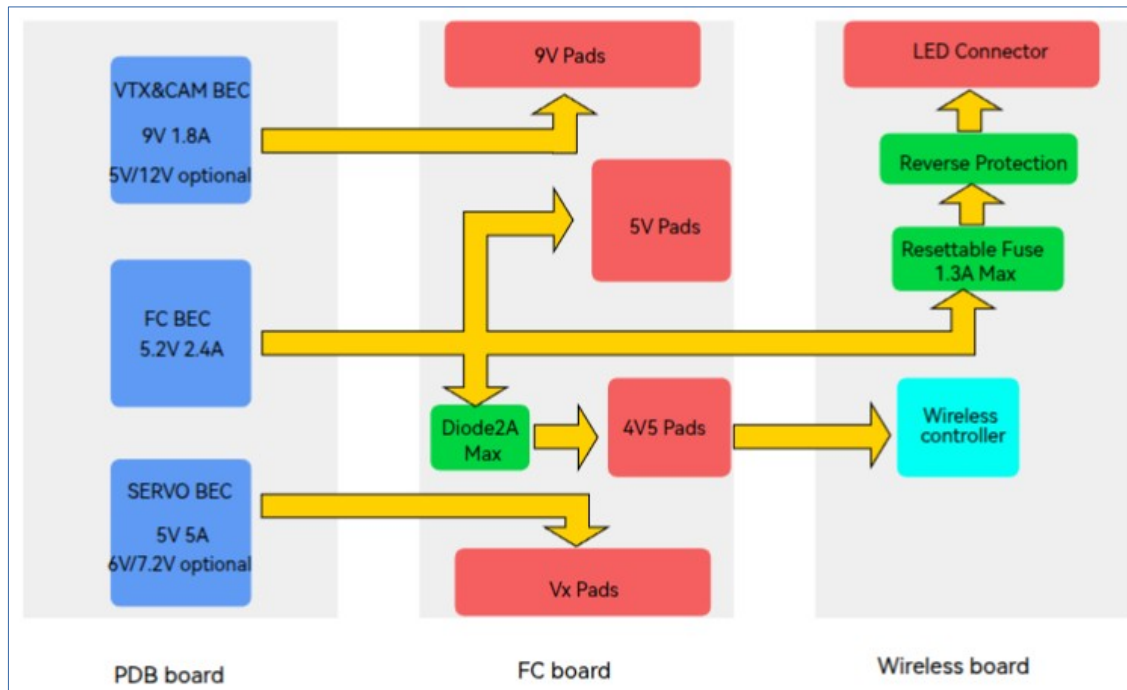
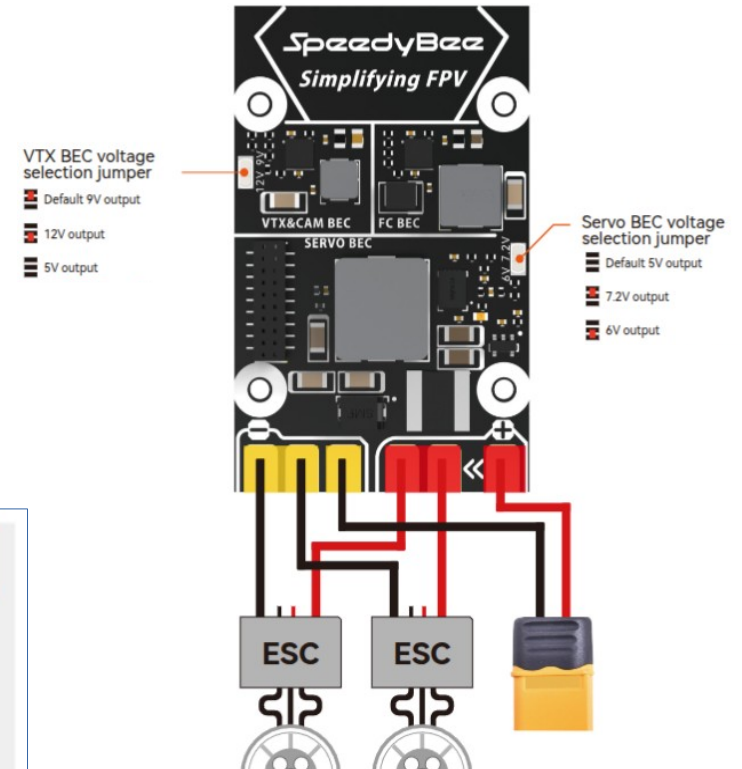
TPS630701 Buck-Boost Converter



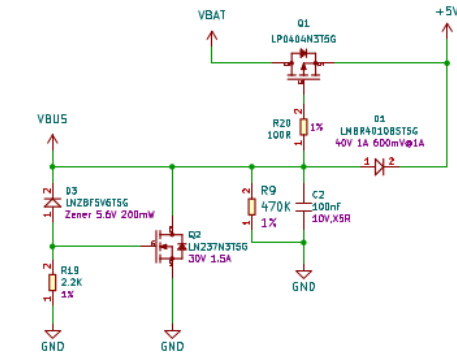
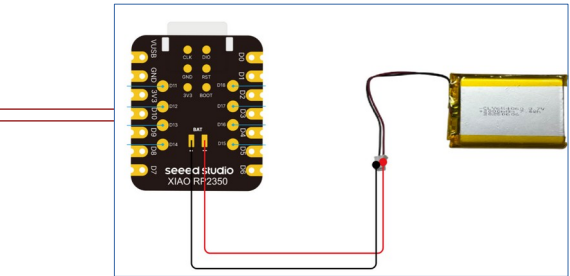
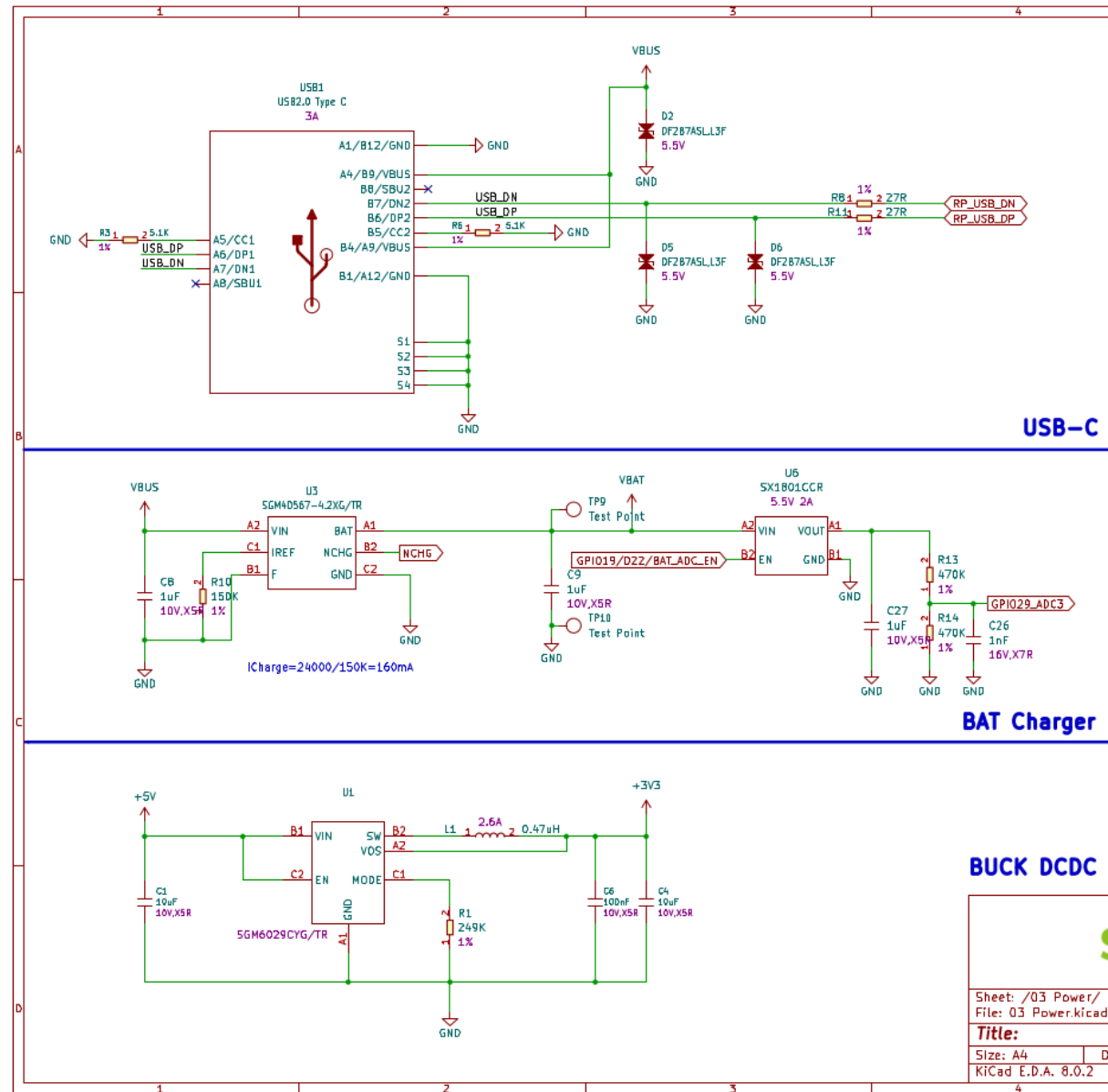
TPS630701 Buck-Boost Converter



電源架構範例



電源架構範例



BUCK DCDC

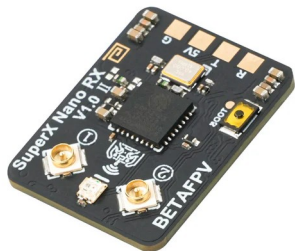
Power Path

seed studio

Sheet: /03 Power/
File: 03 Power.kicad_sch

Title:		CC BY-SA 4.0
Size: A4	Date:	Design:
KiCad E.D.A. 8.0.2	Id: 3/5	Rev:

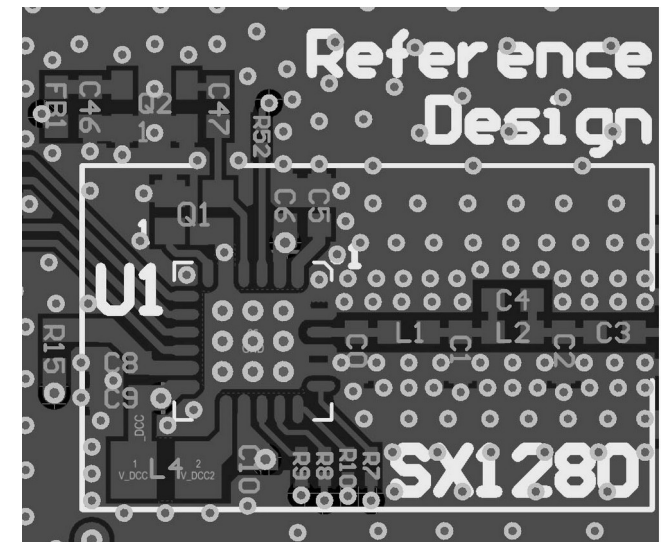
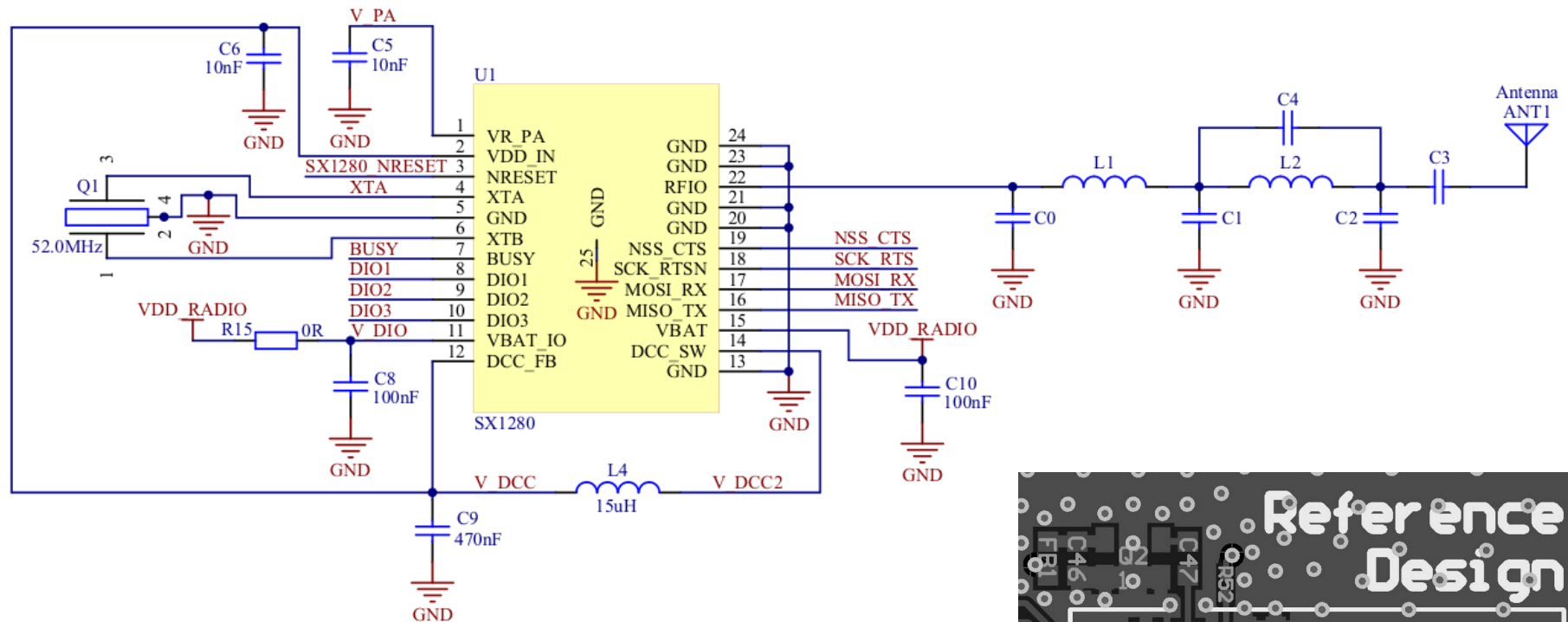
ELRS Receiver



	<u>SuperD RX</u>	<u>Micro RX</u>	<u>Nano RX</u>	<u>Lite RX</u>
Weight	1.1g (RX only)	3.5g (RX only)	0.7g (RX only)	0.46g (Flat V1.2)
Antenna	U.FL + T antenna	U.FL + T antenna	U.FL + T antenna	SMD ceramic antenna
MCU	ESP32-PICO-D4, dual SX1280	ESP8285	ESP8285	ESP8285
RF Chip	Dual SX1280 + Dual PA	SX128X	SX1280 + PA, SX1276 + PA	SX128X
Frequency	2.4GHz ISM	2.4GHz ISM	2.4GHz ISM, 915MHz (FCC), 868MHz (EU)	2.4GHz ISM
Telemetry Power	100mW	17mW	100mW	17mW
Serial Output Protocol	CRSF	PWM or CRSF	CRSF	CRSF
Model Type	FPV Drone, Fixed-wing aircraft	Fixed-wing aircraft, Helicopter, RC Cars, RC Boats	FPV Drone	FPV Drone, Whoop Drone

ELRS Receiver - Reference Design

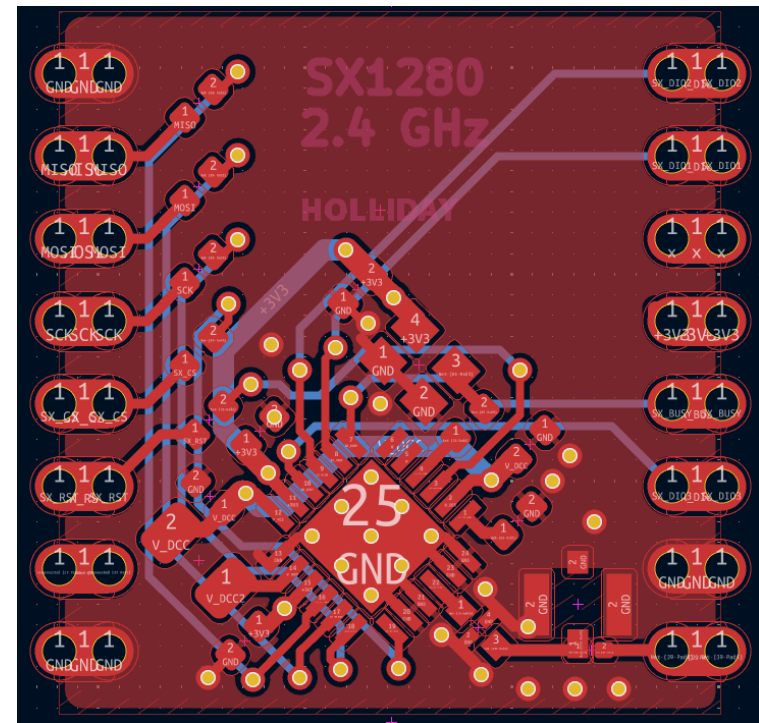
The long range 2.4 GHz application circuit is shown below:



ELRS Receiver - Reference Design BOM

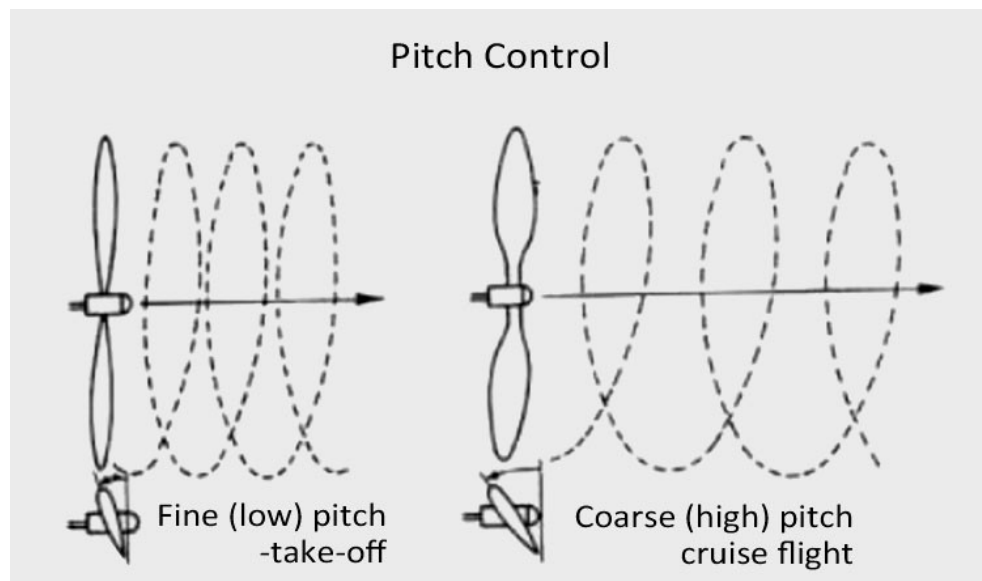
RefDes	MPN	Geom	Value	Description
U1	SX1280	VQFN24 4x4mm	SX1280	SX1280 2.4 GHz High Link Budget Transceiver with LoRa® technology
R15	CRCW04020000Z0ED	0402	0 Ω	Thick Film Resistor $\pm 1\%$, 1/16W
C0	GRM1555C1HR80BA01D	0402	0.8 pF	Multilayer ceramic capacitors C0G ± 0.1 pF, 50 V
C1	GRM1555C1H1R2BA01D	0402	1.2 pF	Multilayer ceramic capacitors C0G ± 0.1 pF, 50 V
C2	GRM1555C1H1R2BA01D	0402	1.2 pF	Multilayer ceramic capacitors C0G ± 0.1 pF, 50 V
C3	GRM1555C1H101JA01D	0402	100 pF	Multilayer ceramic capacitors C0G $\pm 5\%$, 50 V
C4	GRM1555C1HR50WA01D	0402	0.5 pF	Multilayer ceramic capacitors C0G ± 0.05 pF, 50 V
C5	GRM155R71E103KA01D	0402	10 nF	Multilayer ceramic capacitors X7R $\pm 10\%$, 25 V
C6	GRM155R71E103KA01D	0402	10 nF	Multilayer ceramic capacitors X7R $\pm 10\%$, 25 V
C8	GRM155R71C104KA88D	0402	100 nF	Multilayer ceramic capacitors X7R $\pm 10\%$, 16 V
C9	GRM155R61A474KE15D	0402	470 nF	Multilayer ceramic capacitors X5R $\pm 10\%$, 10 V
C10	GRM155R71C104KA88D	0402	100 nF	Multilayer ceramic capacitors X7R $\pm 10\%$, 16 V
L1	LQW15AN3N0B80D	0402	3.0 nH	Wire-wound Inductor ± 0.1 nH
L2	LQW15AN2N5C00D	0402	2.5 nH	Wire-wound Inductor ± 0.2 nH
L4	MLZ2012M150W	0805	15 μ H	MLZ2012 Multilayer Shielded Inductor $\pm 5\%$
Q1	NX2016SA-52.000000MHZ	NX2016SA	52.000 MHz	Crystal unit, NDK Ref: EXS00A-CS07103, Tol. ± 10 ppm, Cload = 10 pF

Miniature 2.45GHz Impedance Matched Front-End Filter Optimized for Semtech SX1280

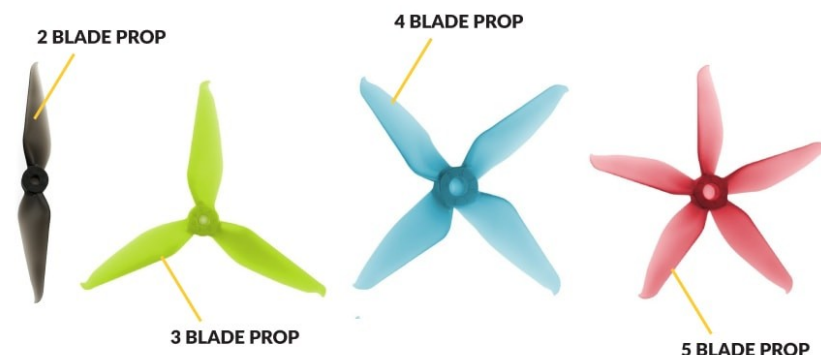


<https://github.com/maholli/sx1280-breakout>

四軸無人機 槳葉 (Diameter, Pitch, Blade)



High pitch 槳葉等效面積較大，在相同轉速下，可以提供較大推力，等同於變速腳踏車的高速檔。**Low pitch** 低轉速時提供較大扭力，速度改變反應也較快。



槳葉數愈多，飛行愈平穩，較不易受周遭氣流影響，操控性更佳。但增加的 Drag，也會導致效率降低，功耗較大。轉換公式如下：

$$D * D * D * P * \sqrt{(N-1)}$$

75mm, 2 blade, P=0.9in



火箭鳥
自造人倉庫

RB — Makers' Depot

四軸無人機 電池

C Rating	Time
30C	2 mins
20C	3 mins
10C	6 mins
5C	12 mins
2C	30 mins
1C	1 hour
0.5C or C/2	2 hours
0.2C or C/5	5 hours
0.1C or C/10	10 hours
0.05C or C/20	20 hours

C rating 告訴我們電池在最大供應電流使用下，可以用多久。

2C RATE EXAMPLE

2300mAh Battery

$$2300\text{mAh} / 1000 = 2.3\text{A}$$

$$2\text{C} \times 2.3\text{A} = 4.6\text{A available}$$

$$1 / 2\text{C} = 0.5 \text{ hours}$$

$$60 / 2\text{C} = 30 \text{ minutes}$$



計算最大供應電流

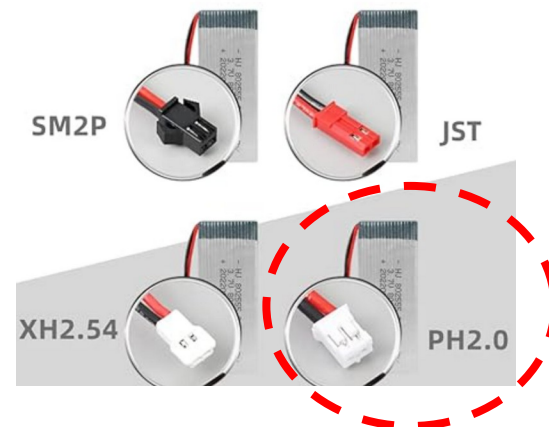
30C RATE EXAMPLE

2300mAh Battery

$$2300\text{mAh} / 1000 = 2.3\text{A}$$

$$30\text{C} \times 2.3\text{A} = 69\text{A available}$$

$$60 / 30\text{C} = 2 \text{ minutes}$$



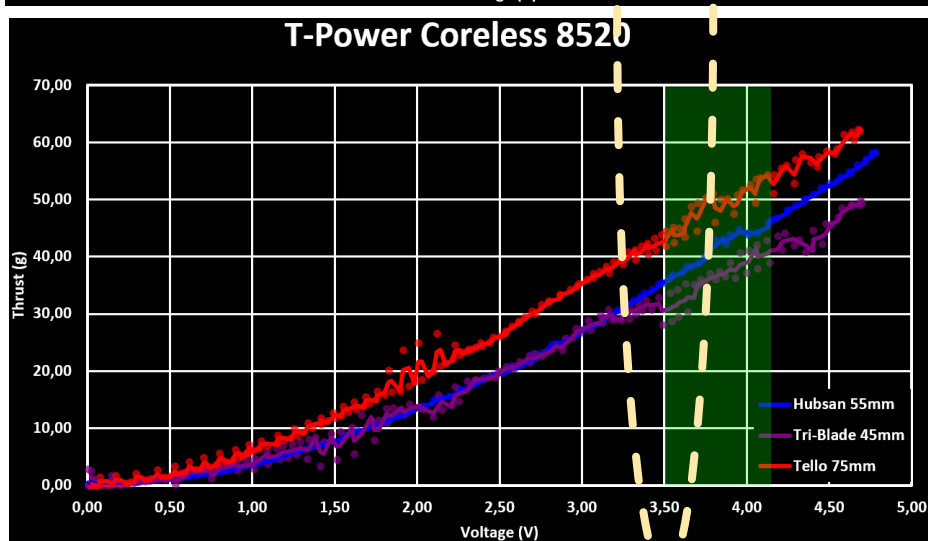
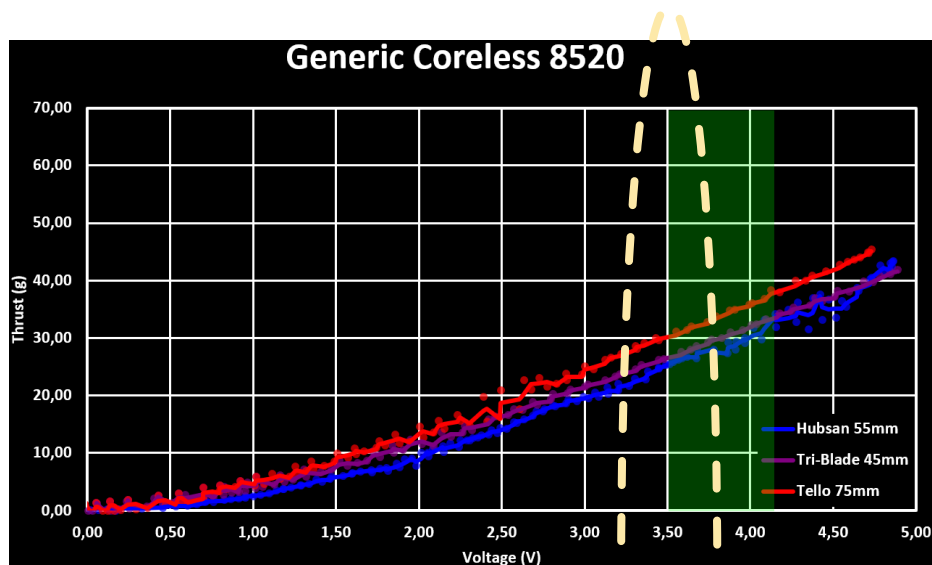
接頭有四種



火箭鳥
自造人倉庫

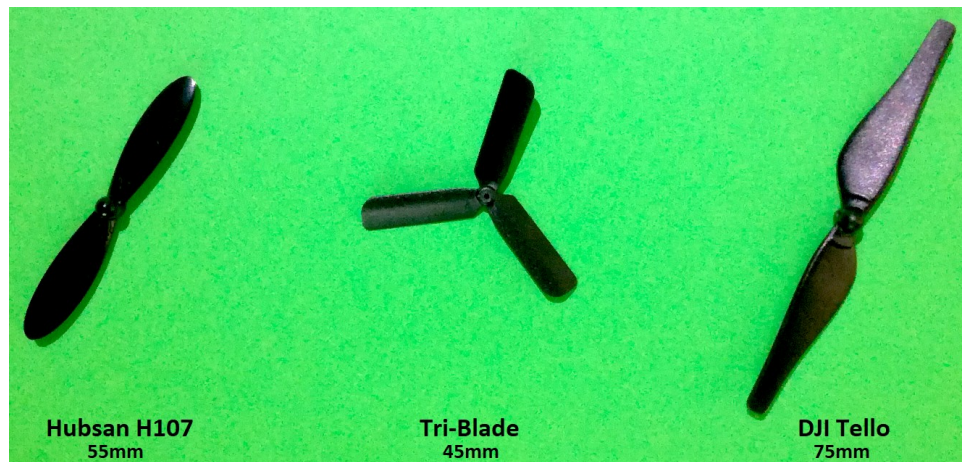
RB — Makers' Depot

四軸無人機 馬達 與 槳葉



<https://notblackmagic.com/projects/motor-test-stand/>

槳葉推力與電壓的關係



同樣的馬達型號，使用相同的槳葉，不同馬達製造商會有不同的推力表現。



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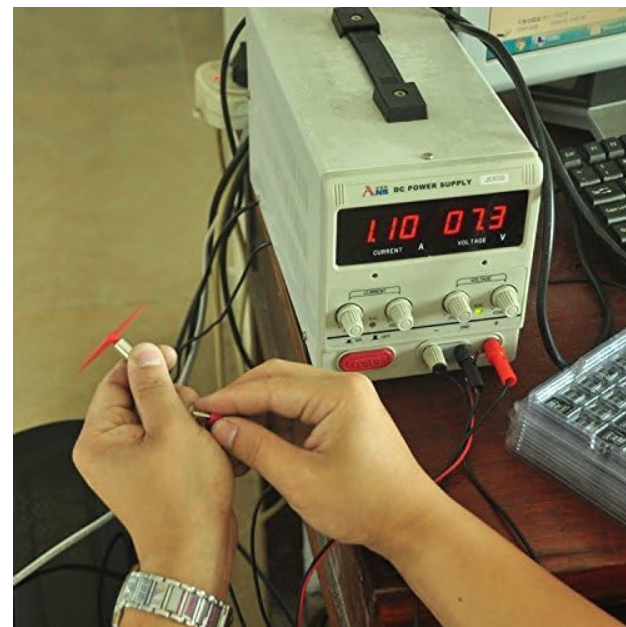
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四軸無人機 馬達 與 電池

Specifications:

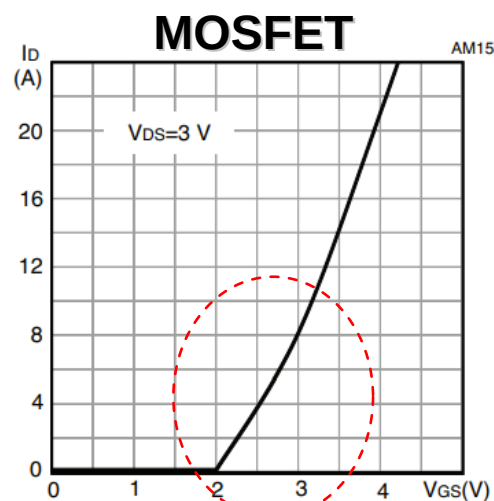
- Size : 8.5mm diameter x 20mm length
- Motor Type : Brushed Coreless
- Voltage: 3.7-5VDC
- RPM(no load) : 39000
- Rated load current: 1750mA max
- Max Current : 2 Amps
- Propeller Size : 75mm Diameter
- Shaft diameter : 1mm
- Weight with propeller : 5 grams

<https://abra-electronics.com/quadcopters/motors/d-m-8520-8520-coreless-motor-reverse.html>



800 mAh, 25C
 $0.8 \text{ A} \times 25 = 20 \text{ A}$
 $20 \text{ A} / 4 = 5 \text{ A}$

600 mAh, 25C
 $0.6 \text{ A} \times 25 = 15 \text{ A}$
 $15 \text{ A} / 4 = 3.75 \text{ A}$



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